

Problem Set 1: Rational Choice, Preferences & Uncertainty

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 2]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with $\langle AI \rangle$ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Expected utility with concrete numbers — 15 points). *An agent has utility $u(x) = \sqrt{x}$ defined over final wealth x (not over gains and losses); initial wealth is zero unless stated otherwise.*

- (a) **(4 pts)** *Lottery L: 50/50 win \$100 or \$0. Compute the expected value, expected utility, certainty equivalent, and risk premium of L. Say in one sentence what the risk premium means behaviorally.*
- (b) **(4 pts)** *The agent now has wealth \$100 and faces a loss of \$64 with probability 1/4 (final wealth \$36 if the loss hits). Compute her expected utility and certainty equivalent, and show that her maximum willingness to pay for full insurance is \$19. Compare it to the expected loss and interpret the difference.*
- (c) **(4 pts)** *An insurer offers full coverage at premium \$18. Does she buy? At premium \$20? State the general rule (one inequality in words) for when a risk-averse agent buys full insurance with a loaded premium.*
- (d) **(3 pts)** *A classmate concludes from (a) that “expected utility theory predicts people are risk averse.” Correct them in two sentences: what does EU assume, and where does risk aversion actually come from in the theory?*

Question 2 (The Allais pattern, by hand — 15 points). *Consider the two questions from Lecture 1 (Kahneman & Tversky 1979 payoffs):*

- **Q1:** $A = \$2400$ for sure; $B = 33\% \$2500, 66\% \$2400, 1\% \$0$.
- **Q2:** $C = 34\% \$2400, 66\% \0 ; $D = 33\% \$2500, 67\% \0 .

In KT’s data, 82% chose A and 83% chose D (the same subjects answered both questions).

- (a) **(6 pts)** *Normalize $u(0) = 0$. Write the expected-utility inequality for $A \succ B$, simplify it, and show it is the same inequality as $C \succ D$. Conclude that no utility function rationalizes choosing both A and D. Hint: cancel the common $0.66 u(2400)$ term.*
- (b) **(3 pts)** *Show that at least 65% of KT’s subjects must have chosen the pair (A, D). Hint: what is the smallest possible overlap between the 82% and the 83%?*

- (c) (**3 pts**) State in one or two sentences which axiom this pattern violates and what the two questions have in common that the axiom says must not matter.
- (d) (**3 pts**) A referee writes: “the Allais pattern is just decision noise.” Give the one-sentence rebuttal that uses the direction of the violation.

Part B: Experimental Design (45 points)

Question 3 (Does the certainty effect survive real incentives?). *KT’s 82/83 result used hypothetical high stakes. Design an experiment that tests whether the certainty effect survives real incentives, and how it moves with stake size. Specify:*

- (a) (**6 pts**) Subject pool and recruitment, and why this population is appropriate for the question.
- (b) (**12 pts**) Treatments. Your design must cross incentive realness (hypothetical vs. paid) with at least two stake levels, and you must say what each comparison identifies. Explain how you keep the probabilities and payoff ratios of the KT questions intact while scaling the stakes.
- (c) (**9 pts**) The payment mechanism. If you pay one randomly selected decision per subject, explain the incentive-compatibility argument — and the standard objection to it in this particular experiment (Lecture 1 flagged it), plus one design choice that mitigates the objection.
- (d) (**9 pts**) Hypotheses and rough sample-size reasoning: state the violation rate you expect under each treatment (anchor on KT’s 65% floor and on any published real-stakes replication you know; a reasoned guess with a stated source is fine), and the approximate n per cell needed to detect the hypothetical-vs-real difference you posit. Simulation-based reasoning is welcome; formulas are not required.
- (e) (**9 pts**) The biggest confound and your design answer to it.

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The suspiciously perfect replication). You run `module1-simulation-lab.ipynb` with the baseline persona and the original KT payoffs. Across the 30 sampled responses per cell (at the lab’s fixed temperature), your agents choose A 82% of the time and D 83% of the time — matching Kahneman & Tversky’s published shares to the percentage point. Your lab partner is thrilled: “the model has human risk preferences!”

- (a) (**8 pts**) Give two mechanisms that could produce this match, at least one of which implies your partner is wrong, and say what each mechanism predicts the agents would do on a new choice problem with the same structure but unfamiliar payoffs.
- (b) (**9 pts**) Design the diagnostic simulation that separates your two mechanisms using only changes to the prompt or the persona grid (e.g. novel payoff numbers, disguised framing, paraphrase, the “you are an economist” persona). State the predicted result pattern under each mechanism. Run it if API budget allows and report; otherwise state the predictions precisely.

- (c) **(8 pts)** Suppose the diagnostic shows the match is recitation, not behavior. Write the two-sentence “limits of silicon subjects” paragraph you would put in a paper that nevertheless uses this lab as a pilot — what CAN the pilot still establish, and what must wait for human subjects?

Attach your AI transcripts and, if you ran code, the results CSV.